ELEC3104: Digital Signal Processing

Topic 3: Digital Filter Fundamentals

Model Answers

Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2026

Topic 3: Pass-Level Tutorial Questions

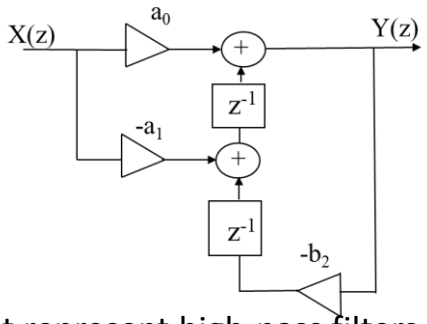
1. A digital filter structure is shown below. Calculate the dc gain of the filter, in terms of the filter coefficients. (Tick the correct answer)

   (a) $1 + a_0 - a_1$

   (b) $\frac{a_0 - a_1}{1 + b_2}$

   (c) $\frac{a_0 - a_1}{1 - b_2}$

   (d) Non of the above



$$H(z) = \frac{a_0 - a_1 z^{-1}}{1 + b_2 z^{-2}} \quad \therefore \quad H(\theta) = \frac{a_0 - a_1 e^{-j\theta}}{1 + b_2 e^{-j2\theta}}$$

$$\text{DC gain} = H(\theta)|_{\theta=0} = \frac{a_0 - a_1}{1 + b_2}$$

2. Tick all transfer functions that represent high-pass filters, where $0 < \alpha < 1$.

   $(a)\ \ H(z) = 1 - \alpha\, z^{-1}$ $\qquad (b)\ \ H(z) = \frac{1}{1 - \alpha\, z^{-1}}$

   $(c)\ \ H(z) = 1 + \alpha\, z^{-1}$ $\qquad (d)\ \ H(z) = \frac{1}{1 + \alpha\, z^{-1}}$

(a) and (d) are high-pass filters – see lecture notes (Topic 2)

3. The Digital Filter $H(z) = \frac{2 + 2z^{-1}}{3 - z^{-1}}$; (Tick the correct answer)

   a) is a low-pass filter
   b) is a high-pass filter
   c) is an all-pass filter

$$H(z) = \frac{2}{3}\left(\frac{1 + z^{-1}}{1 - \frac{1}{3}z^{-1}}\right)$$

$1 + z^{-1}$ is a low-pass filter, and $\frac{1}{1 - \alpha z^{-1}}$ is a low-pass filter.

4. A maximum- phase FIR filter has: (Tick the correct answer)

   a) All zeros and poles outside the unit circles
   b) All poles outside the unit circle
   c) All zeros outside the unit circle
   d) All poles and zeros inside the unit circle

Topic 3: Pass-Level Tutorial Questions

5. Which of the following are linear- phase filters: (Tick the correct answer)

(a) $H(z) = 0.25 + 0.5\,z^{-1} + 0.25\,z^{-2}$ (b) $H(z) = z^{-4}$

(c) $H(z) = 1 - z^{-2}$ (d) $H(z) = 1 + 2z^{-1} + 2\,z^{-2} - z^{-3}$

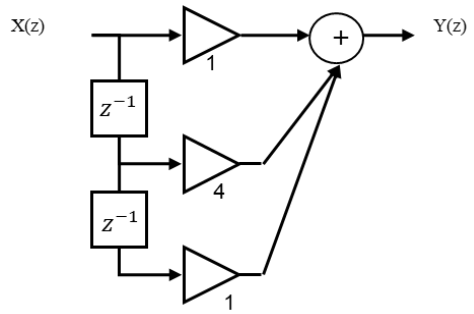
6. What can be said about the following filter ? (Tick the correct answer)

a) It has minimum phase and a finite impulse response.

b) It has linear phase and an infinite impulse response.

c) It has an infinite impulse response.

d) It has linear phase and a finite impulse response



7. Which of the following system is linear, time - invariant, causal and stable? (Tick the correct answer)

a) y[n] = |x[n]|

b) y[n] = x[n] + n x[n+1]

c) y[n] = x[-n]

d) $y[n] = \frac{1}{N}\sum_{k=0}^{N-1} x[n-k]$

(a) & (c) Impulse responses are symmetric about their midpoints

(b) see lecture notes

$$H(z) = 1 + 4z^{-1} + z^{-2} = \frac{z^2 + 4z + 1}{z^2}$$

- **FIR filter:** Therefore, (b) and (c) are wrong.
- **Symmetrical filter coefficients — linear phase.**

$$z = \frac{-4 \pm \sqrt{16 - 4}}{2} = -2 \pm \sqrt{3}$$

One zero is inside the unit circle and one is outside the unit circle.

Therefore, **(a) is wrong.**

a) $y[n] = |x[n]|$ — **Nonlinear**

b) $y[n] = x[n] + n\,x[n+1]$ — **Time-varying**

c) $y[n] = x[-n]$ — **Noncausal**

✓ d) $y[n] = \frac{1}{N}\sum_{k=0}^{N-1} x[n-k]$ — **An FIR filter**

Topic 3: Pass-Level Tutorial Questions

1. Compute the 3-dB bandwidth of the following filters:

$$H_1(z) = \frac{1-a}{1-az^{-1}}; \quad H_2(z) = \frac{1-a}{2}\frac{1+z^{-1}}{1-az^{-1}}, \qquad 0 < a < 1$$

Which filter has a smaller 3dB bandwidth?

$$H_1(\theta) = \frac{1-a}{1-ae^{-j\theta}}$$

$$\begin{aligned} |H_1(\theta)|^2 &= H_1(\theta)\, H_1^*(\theta) \\ &= \frac{1-a}{1-ae^{-j\theta}} \cdot \frac{1-a}{1-ae^{j\theta}} \\ &= \frac{(1-a)^2}{1+a^2-2a\cos\theta} \end{aligned}$$

$$|H_1(\theta)|^2\big|_{\theta=\theta_{c1}} = \frac{1}{2}$$

$$\frac{(1-a)^2}{1+a^2-2a\cos\theta_{c1}} = \frac{1}{2} \quad \Rightarrow \quad \boxed{\cos\theta_{c1} = \frac{4a-a^2-1}{2a}}$$

$$H_2(\theta) = \frac{1-a}{2}\,\frac{1+e^{-j\theta}}{1-ae^{-j\theta}}$$

$$\begin{aligned} |H_2(\theta)|^2 &= \left(\frac{1-a}{2}\right)^2 \frac{(1+\cos\theta)^2+\sin^2\theta}{(1-a\cos\theta)^2+a^2\sin^2\theta} \\ &= \left(\frac{1-a}{2}\right)^2 \frac{2+2\cos\theta}{1+a^2-2a\cos\theta} \end{aligned}$$

$$|H_2(\theta)|^2\big|_{\theta=\theta_{c2}} = \frac{1}{2}$$

$$\frac{1}{2} = \left(\frac{1-a}{2}\right)^2 \frac{2+2\cos\theta_{c2}}{1+a^2-2a\cos\theta_{c2}}$$

$$\therefore \quad \boxed{\cos\theta_{c2} = \frac{2a}{1+a^2}}$$

$$\cos\theta_{c2} > \cos\theta_{c1} \qquad \text{for } 0 < a < 1$$

$$\therefore \quad \boxed{\theta_{c2} < \theta_{c1}}$$

Topic 3: Pass-Level Tutorial Questions

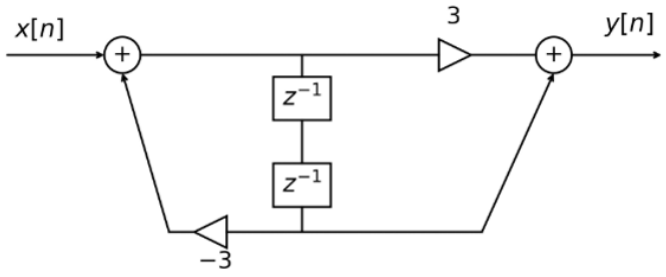
2. Find the magnitude response of the filter *H(z)* and comment on your result. $H(z) = \frac{1+3z^2}{3+z^2}$; Draw the canonic realisation of the filter *H(z)*.

$$H(\theta) = \frac{1 + 3e^{-j2\theta}}{3 + e^{-j2\theta}}$$

$$\begin{aligned}
|H(\theta)|^2 &= H(\theta)\, H^*(\theta) \\
&= \left(\frac{1 + 3e^{-j2\theta}}{3 + e^{-j2\theta}}\right)\left(\frac{1 + 3e^{j2\theta}}{3 + e^{j2\theta}}\right) \\
&= \frac{(1 + 3\cos 2\theta)^2 + (3\sin 2\theta)^2}{(3 + \cos 2\theta)^2 + (\sin 2\theta)^2} \\
&= \frac{10 + 6\cos 2\theta}{10 + 6\cos 2\theta} \\
&= 1
\end{aligned}$$

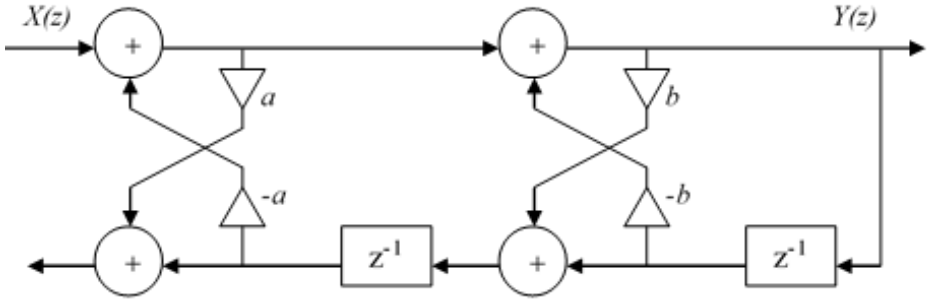
$$\boxed{|H(\theta)| = 1}$$

$$H(z) = \frac{1 + 3z^2}{3 + z^2} = \frac{3 + z^{-2}}{1 + 3z^{-2}}$$



Canonic form

3. Determine the frequency response *H(θ)* of the filter represented by the following structure:



$$y[n] = w[n] - b\,y[n-1] \qquad (1)$$

$$v[n] = y[n-1] + b\,y[n] \qquad (2)$$

$$w[n] = x[n] - a\,v[n-1] \qquad (3)$$

Substitute (3) into (1) to eliminate $w[n]$:

$$y[n] = x[n] - a\,v[n-1] - b\,y[n-1]$$

From (2)

$$v[n-1] = y[n-2] + b\,y[n-1]$$

Therefore

$$y[n] = x[n] - a\,y[n-2] - ab\,y[n-1] - b\,y[n-1]$$

$$\boxed{\frac{Y(z)}{X(z)} = \frac{1}{1 + b(a+1)z^{-1} + az^{-2}}}$$

$$\therefore \quad \boxed{H(\theta) = \frac{1}{1 + b(a+1)e^{-j\theta} + ae^{-j2\theta}}}$$

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4. A two-pole lowpass filter has the transfer function

$$H(z) = \frac{k_0}{(1 - b_1 z^{-1})^2}$$

Determine the values of k_0 and b_1 such that the frequency response $H(\theta)$ satisfy the conditions

$$H(0) = 1; \quad |H(\theta)|^2_{\theta=\frac{\pi}{4}} = \frac{1}{2}$$

$$H(\theta) = \frac{K_0}{(1 - b_1 e^{-j\theta})^2}$$

$$|H(0)| = 1 \quad \Rightarrow \quad 1 = \frac{K_0}{(1 - b_1)^2}$$

$$\therefore \quad \boxed{K_0 = (1 - b_1)^2}$$

$$|H(\theta)|^2 = \frac{K_0}{(1 - b_1 e^{-j\theta})^2} \cdot \frac{K_0}{(1 - b_1 e^{j\theta})^2}$$

$$= \frac{K_0^2}{(1 + b_1^2 - 2b_1 \cos\theta)^2}$$

$$|H(\theta)|^2\big|_{\theta=\pi/4} = \frac{1}{2} = \frac{K_0^2}{\left(1 + b_1^2 - 2b_1 \frac{1}{\sqrt{2}}\right)^2}$$

$$\Rightarrow \quad (1 + b_1^2 - b_1\sqrt{2})^2 = 2K_0^2$$

$$\therefore \quad 1 + b_1^2 - b_1\sqrt{2} = \pm\sqrt{2}(1 - b_1)^2$$

$$\boxed{b_1 \approx 0.32}$$

$$\boxed{K_0 = (1 - 0.32)^2 \approx 0.46}$$

OR

$$\boxed{b_1 = 3.09, \qquad K_0 = 4.37}$$

5. Implement for the following *H(z) using* a parallel structure

$$H(z) = \frac{z^2 - 4z + 3}{z(z + 0.5)^2}$$

$$\frac{z^2 - 4z + 3}{z(z + 0.5)^2} = \frac{A}{z} + \frac{Bz + C}{z^2 + z + \frac{1}{4}}$$

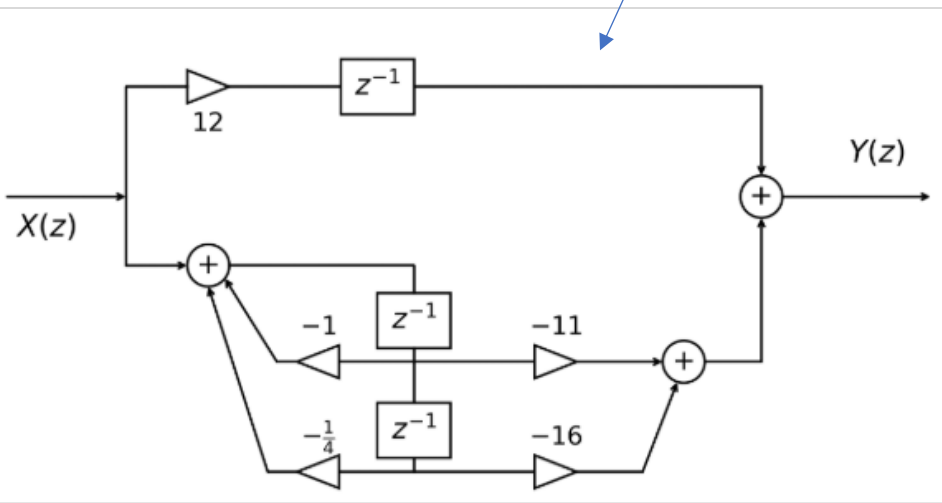
$$z^2 - 4z + 3 = Az^2 + Az + \frac{A}{4} + Bz^2 + Cz$$

By equating the coefficients, we obtain:

$$A + B = 1, \qquad A + C = -4, \qquad \frac{A}{4} = 3$$

$$\therefore \quad \boxed{B = -11, \qquad C = -16, \qquad A = 12}$$

$$H(z) = 12z^{-1} + \frac{-11z^{-1} - 16z^{-2}}{1 + z^{-1} + \frac{1}{4}z^{-2}}$$



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6. A notch filter is given by $H(z) = \frac{1+z^{-2}}{1+0.8z^{-2}}$. Determine the magnitude response at $\theta = 0, \frac{\pi}{2}, \text{ and } \pi$. Sketch the magnitude response in the interval $-\pi \leq \theta \leq \pi$.

$$H(\theta) = \frac{1 + e^{-j2\theta}}{1 + 0.8e^{-j2\theta}}$$

$$\text{DC gain} = H(\theta)|_{\theta=0} = \frac{2}{1.8}$$

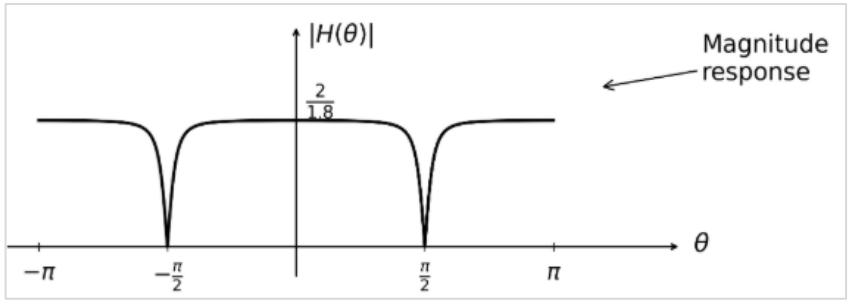
At $\theta = \frac{\pi}{2}$,

$$H(\theta)|_{\theta=\pi/2} = \frac{1 + e^{-j(\pi/2)\times 2}}{1 + 0.8e^{-j(\pi/2)\times 2}}$$

$$= \frac{1 + \cos\pi - j\sin\pi}{1 + 0.8(\cos\pi - j\sin\pi)} = 0$$

At $\theta = \pi$,

$$H(\theta)|_{\theta=\pi} = \frac{1 + e^{-j2\pi}}{1 + 0.8\,(\cos 2\pi - j\sin 2\pi)} = \frac{2}{1.8}.$$



7. Determine and sketch the magnitude and phase responses of the following filters (a) $y[n] = \frac{1}{2}(x[n] - x[n-1])$ (b) $y[n] = x[n] - x[n-8]$; (c) $y[n] = x[n-4]$

(a)

$$Y(z) = \frac{1}{2}X(z) - \frac{1}{2}z^{-1}X(z) \quad \Rightarrow \quad H(z) = \frac{1}{2}(1 - z^{-1})$$

$$H(\theta) = \frac{1}{2}(1 - e^{-j\theta})$$

$$= \frac{1}{2}e^{-j\theta/2}\left(\frac{e^{j\theta/2} - e^{-j\theta/2}}{2j}\right)(2j)$$

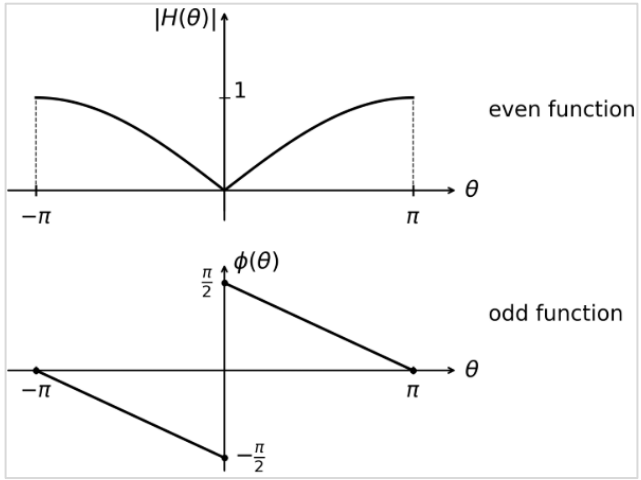
$$= j\,e^{-j\theta/2}\sin\frac{\theta}{2}$$

$$= e^{j(\pi/2-\theta/2)}\sin\frac{\theta}{2}.$$

$$|H(\theta)| = \left|\sin\frac{\theta}{2}\right|$$

$$B(\theta) = \sin\frac{\theta}{2}.$$

$$\phi(\theta) = \begin{cases} \frac{\pi}{2} - \frac{\theta}{2}, & B(\theta) \geq 0, \\ \left(\frac{\pi}{2} - \frac{\theta}{2}\right) \pm \pi, & B(\theta) < 0, \end{cases}$$



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7. Determine and sketch the magnitude and phase responses of the following filters

(a) $y[n] = \frac{1}{2}(x[n] - x[n-1])$ (b) $y[n] = x[n] - x[n-8]$; $(c)\ y[n] = x[n-4]$

(b)

$$y[n] = x[n] - x[n-8]$$

$$H(z) = \frac{Y(z)}{X(z)} = 1 - z^{-8}$$

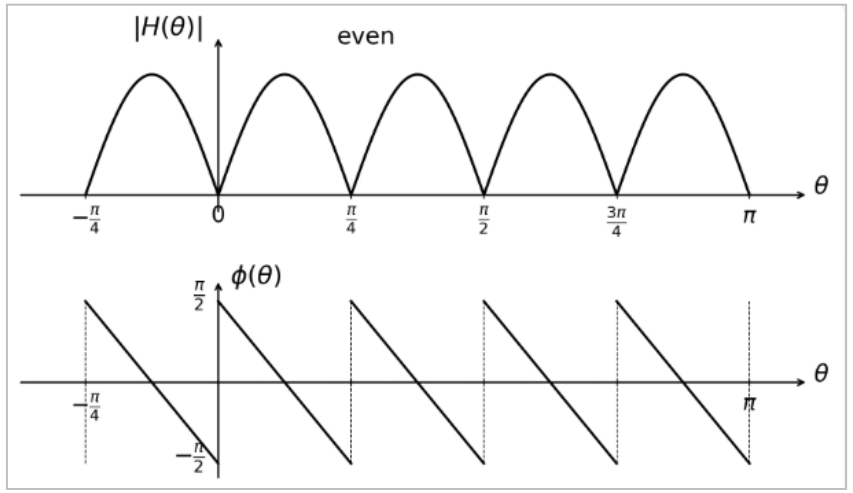
$$H(\theta) = 1 - e^{-j8\theta} = e^{-j4\theta}\left[e^{j4\theta} - e^{-j4\theta}\right]$$

$$H(\theta) = e^{-j4\theta}(2j)\sin 4\theta$$

$$H(\theta) = e^{j(\pi/2 - 4\theta)} \underbrace{2\sin 4\theta}_{B(\theta)}$$

$$|H(\theta)| = |2\sin 4\theta|$$

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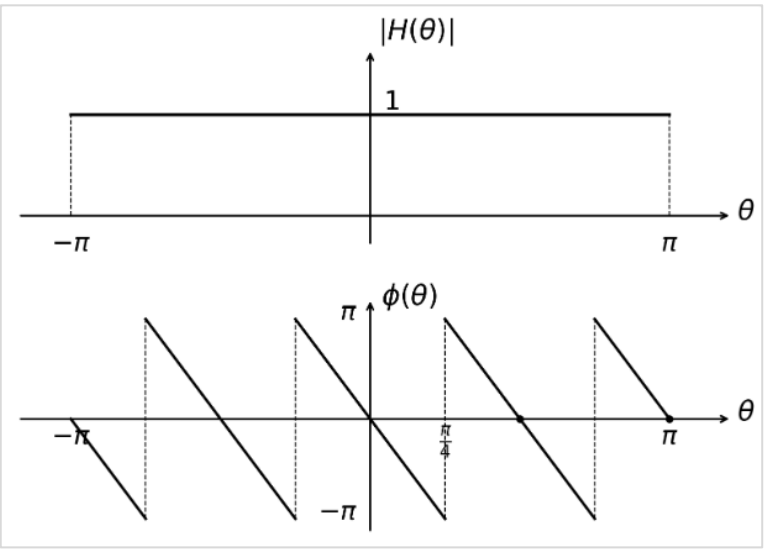
(c)

$$Y(z) = z^{-4}X(z)$$

$$H(z) = z^{-4}; \qquad H(\theta) = e^{-j4\theta}$$

$$|H(\theta)| = 1$$

$$\phi(\theta) = -4\theta$$



Topic 3: Pass-Level Tutorial Questions

1. Compute the 3-dB bandwidth of the following filters:

$$H_1(z) = \frac{1-a}{1-az^{-1}}; \quad H_2(z) = \frac{1-a}{2}\frac{1+z^{-1}}{1-az^{-1}}, \qquad 0 < a < 1$$

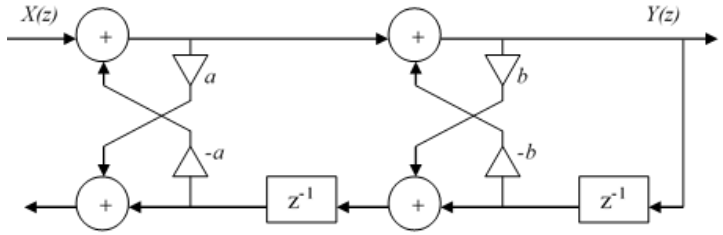
Which filter has a smaller 3dB bandwidth?

2. Find the magnitude response of the filter *H(z)* and comment on your result.

$$H(z) = \frac{1+3z^2}{3+z^2}$$

Draw the canonic realisation of the filter *H(z)*.

3. Determine the frequency response $H(\theta)$ of the filter represented by the following structure:



4. A two-pole lowpass filter has the transfer function

$$H(z) = \frac{k_0}{(1-b_1z^{-1})^2}$$

Determine the values of k_0 and b_1 such that the frequency response $H(\theta)$ satisfy the conditions

$$H(0) = 1; \quad |H(\theta)|^2_{\theta=\frac{\pi}{4}} = \frac{1}{2}$$

5. Implement for the following *H(z) using* a parallel structure"

$$H(z) = \frac{z^2-4z+3}{z(z+0.5)^2}$$

6. A notch filter is given by $H(z) = \frac{1+z^{-2}}{1+0.8z^{-2}}$. Determine the magnitude response at $\theta = 0, \frac{\pi}{2}, and\ \pi$. Sketch the magnitude response in the interval $-\pi \le \theta \le \pi$.

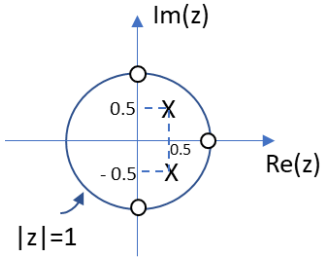
7. Determine and sketch the magnitude and phase responses of the following filters

(a) $y[n] = \frac{1}{2}(x[n] - x[n-1])$ (b) $y[n] = x[n] - x[n-8]$

(c) $y[n] = x[n-4]$

8 (a) Determine the transfer function H(z) of a discrete - time system with the pole-zero map shown below:

(b) Using the pole-zero map , sketch the approximate magnitude response $|H(\theta)|$.



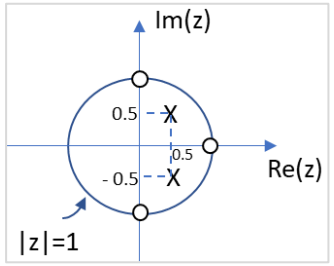
9. A finite impulse filter has seven filter coefficients b_0 to b_6, with $b_0 = 1$, $b_1 = 2$, $b_2 = 3$ and $b_3 = 4$. Determine b_4 to b_6 such that the filter has linear phase.

10. Implement the following transfer function using a cascade structure, direct form II and in parallel structure. Assume all coefficients are real and $a \neq -b$.

$$H(z) = \frac{1}{(1+az^{-1})(1-bz^{-1})}$$

Topic 3: Pass-Level Tutorial Questions

8 (a) Determine the transfer function H(z) of a discrete - time system with the pole-zero map shown below: (b) Using the pole-zero map , sketch the approximate magnitude response $|H(\theta)|$.



Angle of the pole $= \theta$:

$$r = \sqrt{(0.5)^2 + (0.5)^2} = \frac{1}{\sqrt{2}}$$

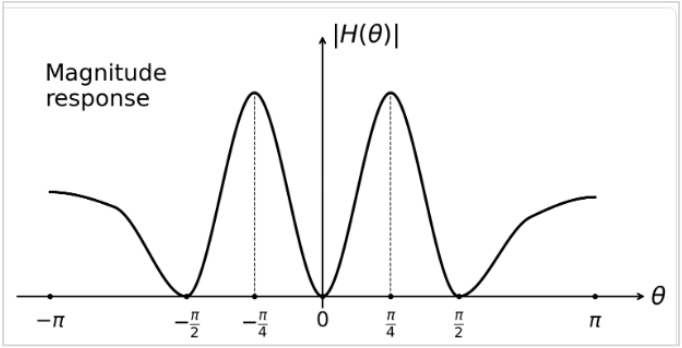
Angle of the pole $= \theta$:

$$\tan\theta = \frac{0.5}{0.5} = 1 \qquad \Rightarrow \qquad \theta = \frac{\pi}{4}$$

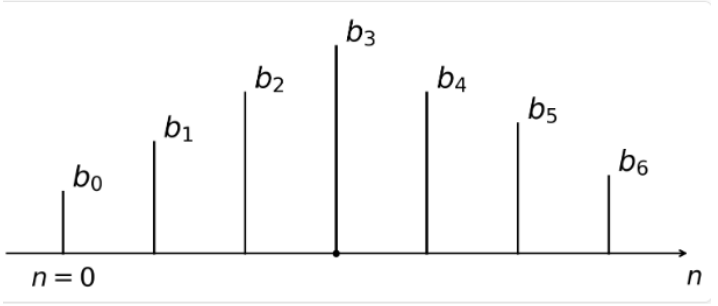
$$H(z) = \frac{(z-1)(z+j)(z-j)}{\left(z - re^{-j\pi/4}\right)\left(z - re^{j\pi/4}\right)}$$

$$= \frac{(z-1)(z^2+1)}{z^2 - \frac{2}{\sqrt{2}}\cos\frac{\pi}{4}\, z + \frac{1}{2}}$$

$$\boxed{H(z) = \frac{(z-1)(z^2+1)}{z^2 - z + \frac{1}{2}}}$$



9. A finite impulse filter has seven filter coefficients b_0 to b_6, with $b_0 = 1$, $b_1 = 2$, $b_2 = 3$ and $b_3 = 4$. Determine b_4 to b_6 such that the filter has linear phase.



The coefficients must be symmetrical about \(b_3\) for a linear-phase filter. Therefore,

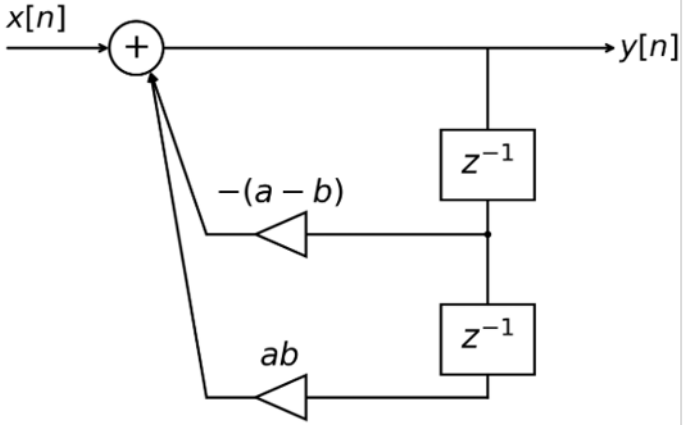
$$\boxed{\begin{aligned} b_4 &= b_2 = 3, \\ b_5 &= b_1 = 2, \\ b_6 &= b_0 = 1. \end{aligned}}$$

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10. Implement the following transfer function using a cascade structure, direct form II and in parallel structure. Assume all coefficients are real and $a \neq -b$; $H(z) = \frac{1}{(1+az^{-1})(1-bz^{-1})}$

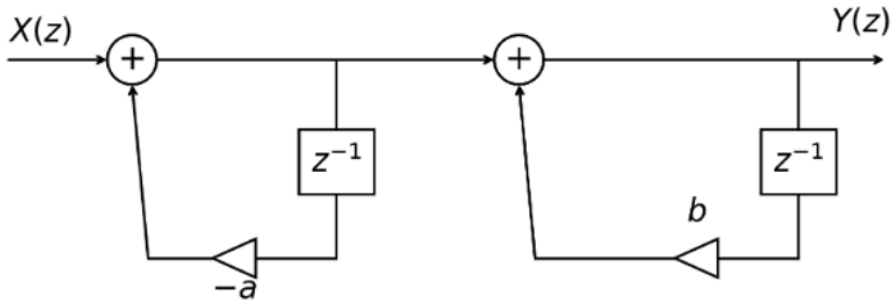
Direct Form II

$$H(z) = \frac{1}{1 + (a-b)z^{-1} - abz^{-2}}$$



Cascade

$$H(z) = \frac{1}{(1 + az^{-1})(1 - bz^{-1})}$$



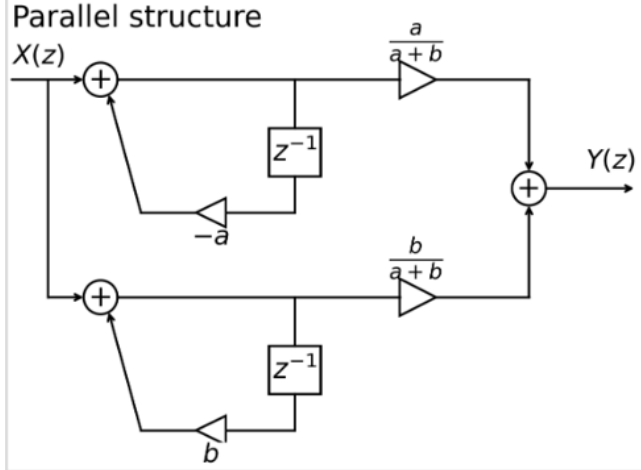
Parallel structure

$$H(z) = \frac{1}{(1 + az^{-1})(1 - bz^{-1})}$$

$$= \frac{A}{1 + az^{-1}} + \frac{B}{1 - bz^{-1}}$$

$$H(z) = \frac{\frac{a}{a+b}}{1 + az^{-1}} + \frac{\frac{b}{a+b}}{1 - bz^{-1}}$$

Parallel structure



Topic 3: Pass-Level Tutorial questions

11. Show that both digital filters given below have the same magnitude response:

(a) $y[n] = \sum_{i=-m}^{m} c_i\, x[n-i]$ (b) $y[n] = \sum_{i=-m}^{m} c_i\, x[n-m-i]$

where $y[n]$ is the output, $x[n]$ is the input and c_i are the coefficients.

$$Y(z) = \sum_{i=-m}^{m} C_i\, X(z)\, z^{-i}$$

$$H(z) = \frac{Y(z)}{X(z)} = \sum_{i=-m}^{m} C_i\, z^{-i}$$

$$\boxed{|H(\theta)| = \left| \sum_{i=-m}^{m} C_i\, e^{-j\theta i} \right|}$$

$$Y(z) = \sum_{i=-m}^{m} C_i\, X(z)\, z^{-m}\, z^{-i}$$

$$H_2(z) = \frac{Y(z)}{X(z)} = \sum_{i=-m}^{m} C_i\, z^{-m}\, z^{-i}$$

$$H_2(\theta) = \sum_{i=-m}^{m} C_i\, e^{-j\theta i}\, e^{-j\theta m}$$

$$|H_2(\theta)| = \left| \sum_{i=-m}^{m} C_i\, e^{-j\theta i} \right| \underbrace{\left| e^{-j\theta m} \right|}_{1}$$

$$= \left| \sum_{i=-m}^{m} C_i\, e^{-j\theta i} \right|$$

$$\therefore \quad \boxed{|H_1(\theta)| = |H_2(\theta)|}$$

12. Consider the following causal IIR transfer function:

$$H(z) = \frac{2z^3 - 4z^2 + 9}{(z-3)(z^2 + z + 0.5)}$$

Is *H(z)* a stable function? If it is not stable, find a stable transfer function *G(z)* such that $|G(\theta)| = |H(\theta)|$.

Answer:

Since H(z) has a pole at z=3 (outside the unit circle), the given transfer function is unstable. To construct a stable transfer function having the same magnitude response, consider another transfer function:

$$G(z) = H(z) \cdot A(z)$$

Make $A(z)$ an all-pass filter:

$$G(z) = \frac{2z^3 - 4z^2 + 9}{(z^2 + z + 0.5)(z-3)} \cdot \underbrace{\frac{z-3}{1-3z}}_{\text{all-pass filter}}$$

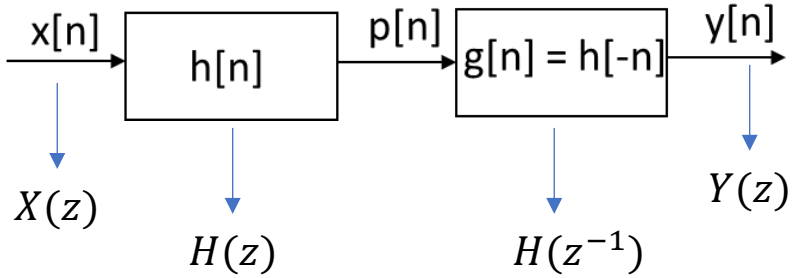
The magnitude response of the all-pass filter is constant, or 1.

$$\therefore \quad |G(\theta)| = |H(\theta)|\,|A(\theta)| \quad \Rightarrow \quad |G(\theta)| = |H(\theta)|$$

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13. Show that for a real valued impulse response $h[n]$ the structure below has a zero − phase frequency Response.

$$H(z) = \left(\frac{1}{2} + z^{-1} + \frac{1}{2}z^{-2}\right)$$



$$\frac{Y(z)}{X(z)} = H(z)\,H(z^{-1}) \quad \text{or} \quad \boxed{\frac{Y(\theta)}{X(\theta)} = H(\theta)\,H^*(\theta)}$$

$$H(\theta) = \frac{1}{2} + e^{-j\theta} + \frac{1}{2}e^{-j2\theta}; \qquad H^*(\theta) = \frac{1}{2} + e^{j\theta} + \frac{1}{2}e^{j2\theta}$$

$$\therefore \quad \frac{Y(\theta)}{X(\theta)} = \left(\frac{1}{2} + e^{-j\theta} + \frac{1}{2}e^{-j2\theta}\right)\left(\frac{1}{2} + e^{j\theta} + \frac{1}{2}e^{j2\theta}\right)$$

$$= \frac{3}{2} + \left(e^{j\theta} + e^{-j\theta}\right) + \frac{1}{4}\left(e^{j2\theta} + e^{-j2\theta}\right).$$

$$\boxed{\frac{Y(\theta)}{X(\theta)} = \frac{3}{2} + 2\cos\theta + \frac{1}{2}\cos 2\theta}$$

Real part only ⇒ Zero phase.

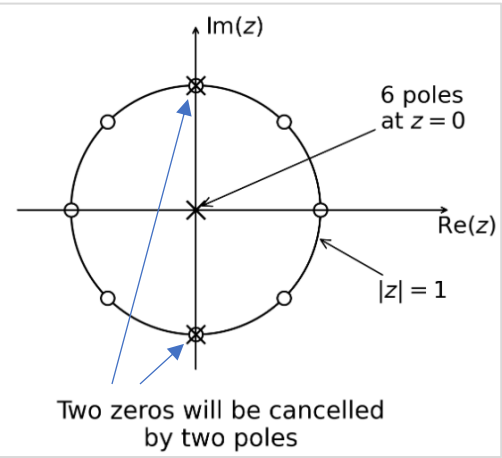
14. The difference equation of a digital filter is given by

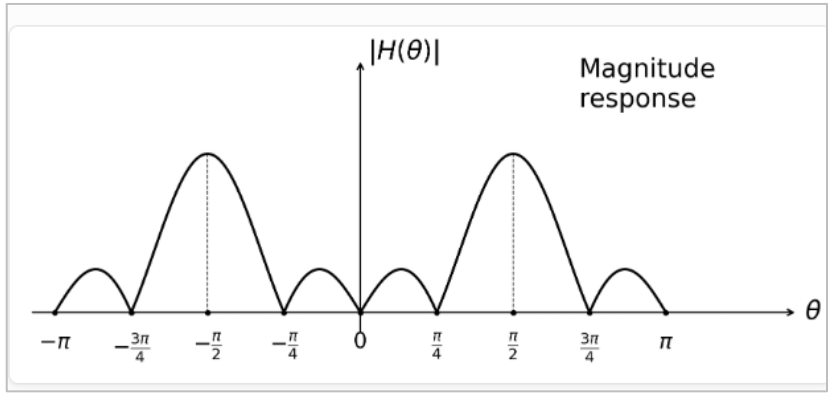
$$y[n] = (1/8)\{x[n] - x(n-8)\} - y(n-2)$$

(a) Find the transfer function for the above filter.

(b) Using the transfer function obtained in part (a) above, draw the pole-zero map for the filter and sketch the corresponding magnitude response.

(a) $$H(z) = \frac{Y(z)}{X(z)} = \frac{1}{8}\,\frac{\overbrace{1 - z^{-8}}^{8\text{ zeros}}}{\underbrace{1 + z^{-2}}_{\text{Two poles}}}$$

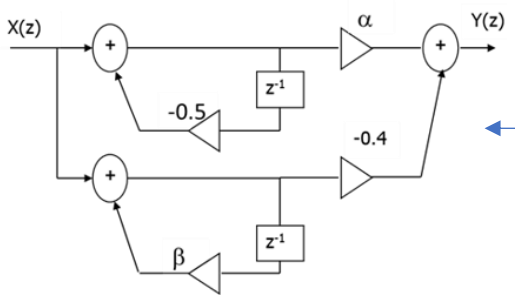




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15. The following figure shows an IIR filter structure, and its transfer function given by $H(z) = \frac{Y(z)}{X(z)} = \frac{3z(5z-2)}{(z+0.5)(3z-1)}$

Determine the values of the multiplier coefficients α and β.



This is a parallel structure

$$H(z) = \frac{Y(z)}{X(z)} = \frac{3z(5z-2)}{(z+0.5)(3z-1)} = \frac{5-2z^{-1}}{(1+0.5z^{-1})(1-\frac{1}{3}z^{-1})}$$

$$\frac{5-2z^{-1}}{(1+\frac{1}{2}z^{-1})(1-\frac{1}{3}z^{-1})} = \frac{A}{1+\frac{1}{2}z^{-1}} + \frac{B}{1-\frac{1}{3}z^{-1}}$$

$$\boxed{\beta = \frac{1}{3}}$$

$$A = \frac{5-2(-2)}{1-\frac{1}{3}(-2)} = \frac{27}{5}$$

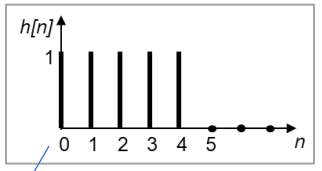
$$B = \frac{5-2(3)}{1+\frac{1}{2}(3)} = -\frac{2}{5}$$

$$\boxed{A = \frac{27}{5}, \quad B = -\frac{2}{5}} \rightarrow \boxed{\alpha = \frac{27}{5}}$$

16. A FIR filter has an impulse response as shown below:

(a) Show that the frequency response of this filter is given by

$$H(\theta) = e^{-j(2\theta)} \frac{sin\left(\frac{5}{2}\theta\right)}{sin\left(\frac{\theta}{2}\right)}$$



(b) Draw an implementation of this filter as a cascade structure.

(a)

$$H(z) = 1 + z^{-1} + z^{-2} + z^{-3} + z^{-4}$$

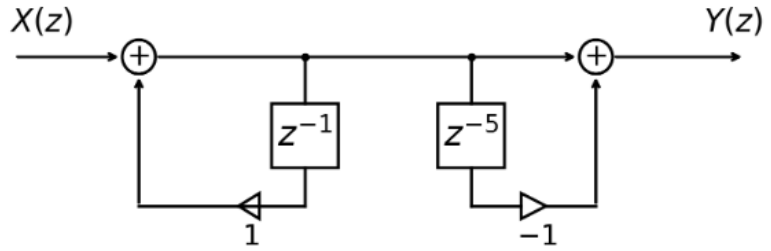
$$H(z) = \frac{1-z^{-5}}{1-z^{-1}}$$

$$H(\theta) = \frac{1-e^{-j5\theta}}{1-e^{-j\theta}} = \frac{e^{-j\frac{5\theta}{2}}\left(e^{j\frac{5\theta}{2}} - e^{-j\frac{5\theta}{2}}\right)}{e^{-j\frac{\theta}{2}}\left(e^{j\frac{\theta}{2}} - e^{-j\frac{\theta}{2}}\right)}$$

$$\boxed{H(\theta) = e^{-j2\theta} \frac{\sin\left(\frac{5\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)}}$$

(b)

$$H(z) = \frac{1-z^{-5}}{1-z^{-1}}$$

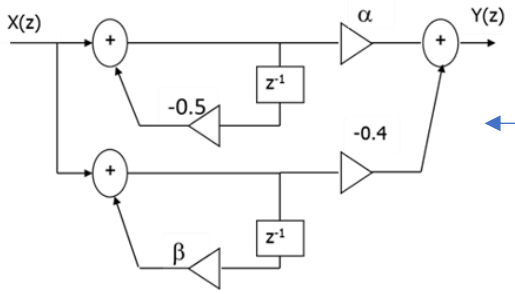


Cascade structure

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15. The following figure shows an IIR filter structure, and its transfer function given by $H(z) = \frac{Y(z)}{X(z)} = \frac{3z(5z-2)}{(z+0.5)(3z-1)}$

Determine the values of the multiplier coefficients α and β.



This is a parallel structure

$$H(z) = \frac{Y(z)}{X(z)} = \frac{3z(5z-2)}{(z+0.5)(3z-1)} = \frac{5-2z^{-1}}{(1+0.5z^{-1})(1-\frac{1}{3}z^{-1})}$$

$$\frac{5-2z^{-1}}{(1+\frac{1}{2}z^{-1})(1-\frac{1}{3}z^{-1})} = \frac{A}{1+\frac{1}{2}z^{-1}} + \frac{B}{1-\frac{1}{3}z^{-1}}$$

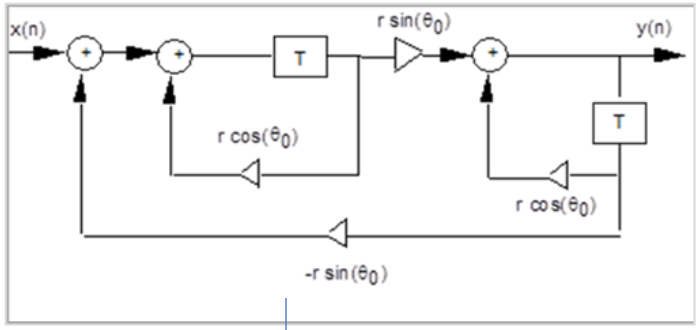
$$\boxed{\beta = \frac{1}{3}}$$

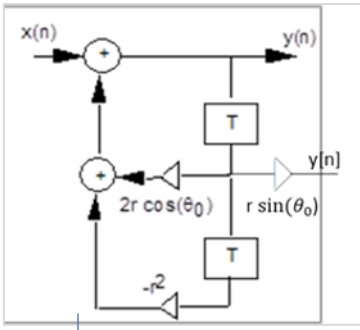
$$A = \frac{5-2(-2)}{1-\frac{1}{3}(-2)} = \frac{27}{5}$$

$$B = \frac{5-2(3)}{1+\frac{1}{2}(3)} = -\frac{2}{5}$$

$$\boxed{\alpha = \frac{27}{5}} \leftarrow \boxed{A = \frac{27}{5}, \quad B = -\frac{2}{5}}$$

17. Show that the systems (i) and (ii) shown below are equivalent:





$$R(z) = X(z) + z^{-1}Y(z)(-r\sin\theta_0) + (r\cos\theta_0)R(z)z^{-1}$$

$$R(z)\left(1 - r\cos\theta_0\, z^{-1}\right) = X(z) - r\cos\theta_0\, z^{-1}Y(z) \quad (1)$$

$$Y(z) = R(z)(r\sin\theta_0)z^{-1} + Y(z)r\cos\theta_0\, z^{-1}$$

$$R(z) = \frac{Y(z)\left[1 - r\cos\theta_0\, z^{-1}\right]}{r\sin\theta_0\, z^{-1}} \quad (2)$$

From (1) and (2),

$$\frac{X(z) - r\sin\theta_0\, z^{-1}Y(z)}{1 - r\cos\theta_0\, z^{-1}} = \frac{Y(z)\left[1 - r\cos\theta_0\, z^{-1}\right]}{r\sin\theta_0\, z^{-1}}$$

$$\boxed{\frac{Y(z)}{X(z)} = \frac{r\sin\theta_0\, z^{-1}}{1 - 2r\cos\theta_0\, z^{-1} + r^2 z^{-2}}}$$

$$\boxed{\frac{Y(z)}{X(z)} = \frac{r\sin\theta_0\, z^{-1}}{1 - 2r\cos\theta_0\, z^{-1} + r^2 z^{-2}}}$$

Both are equivalent.

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18. A third order FIR filter has the transfer function G(z) given by

$$G(z) = (9 + 6z^{-1} - 8z^{-2})(2 + 5z^{-1}).$$

Determine FIR transfer functions having the same magnitude response as G(z) that are:

(a) minimum phase
(b) maximum phase

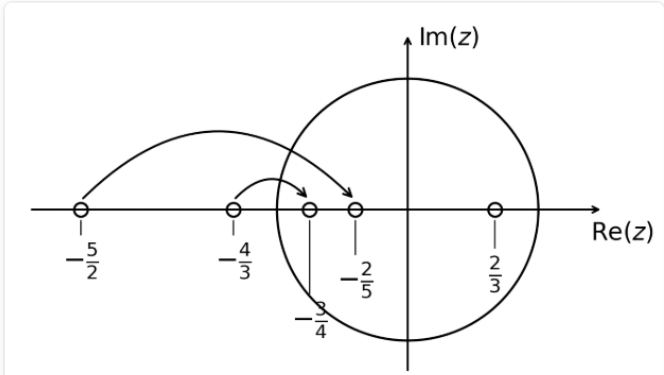
(a)

$$G(z) = (3 - 2z^{-1})(3 + 4z^{-1})(2 + 5z^{-1})$$

$$G(z) = 18 \underbrace{\left(1 - \frac{2}{3}z^{-1}\right)}_{<1} \underbrace{\left(1 + \frac{4}{3}z^{-1}\right)}_{>1} \underbrace{\left(1 + \frac{5}{2}z^{-1}\right)}_{>1}$$

$$G(z) = 60 \underbrace{\left(1 - \frac{2}{3}z^{-1}\right)}_{<1} \underbrace{\left(1 + \frac{3}{4}z^{-1}\right)}_{<1} \underbrace{\left(1 + \frac{2}{5}z^{-1}\right)}_{<1}$$

For minimum phase, reflect the zeros inside the unit circle.

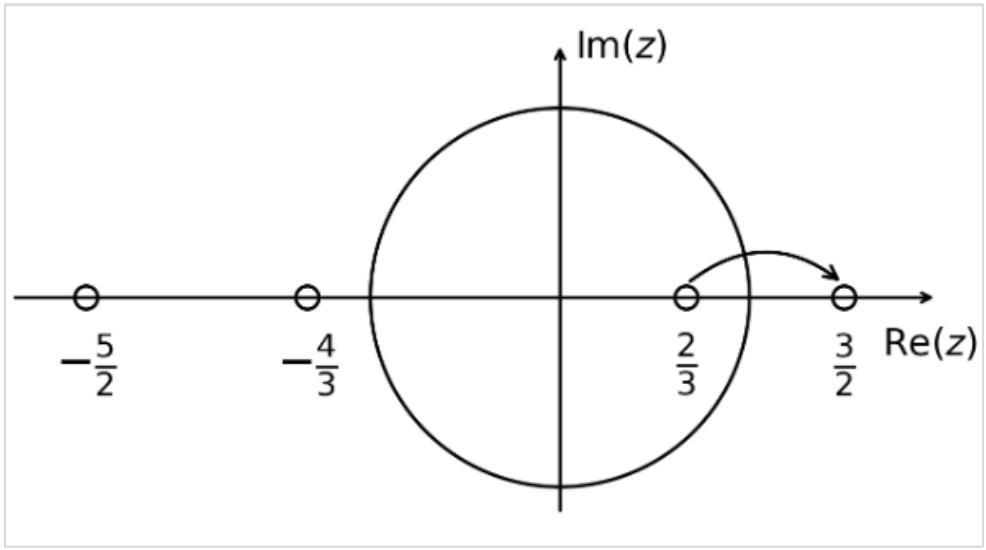


(b)

$$G(z) = -12\left(1 - \frac{3}{2}z^{-1}\right)\left(1 + \frac{4}{3}z^{-1}\right)\left(1 + \frac{5}{2}z^{-1}\right)$$

All the zeros are outside the unit circle.

$\therefore$ $G(z)$ has a maximum-phase response.



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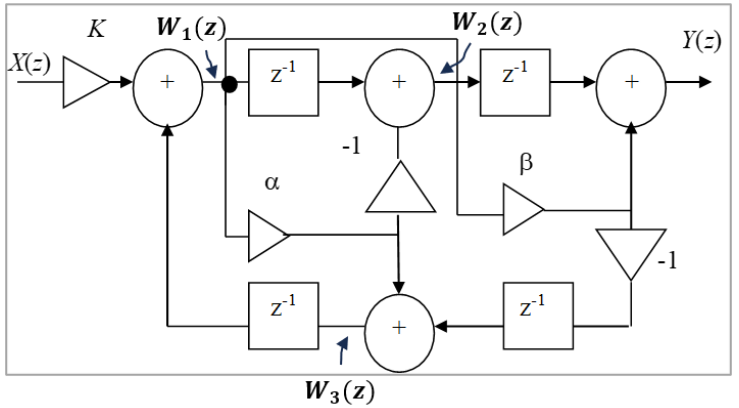
19. Analyse the digital filter structure shown on the right:

(a) Determine its transfer function $H(z) = Y(z)/X(z)$

(b) Determine the value of K such that $H(z)$ has unity gain at $\theta = 0$.

(c) Determine the value of K such that $H(z)$ has unity gain at $\theta = \pi$.

(d) Are the values of K obtained in (b) and (c) different? Explain your answer.



(a)

$$W_1(z) = KX(z) + W_3(z)z^{-1} \quad (1)$$

$$W_2(z) = W_1(z)z^{-1} + (-\alpha)W_1(z) \quad (2)$$

$$W_3(z) = \alpha W_1(z) - \beta W_1(z)z^{-1} \quad (3)$$

$$Y(z) = W_2(z)z^{-1} + \beta W_1(z) \quad (4)$$

From (1) and (3),

$$W_1(z) = KX(z) + z^{-1}\left(\alpha - \beta z^{-1}\right) W_1(z)$$

$$\therefore \quad W_1(z) = \frac{KX(z)}{1 - \alpha z^{-1} + \beta z^{-2}}$$

From (2) and (4),

$$Y(z) = \left[W_1(z)z^{-1} - \alpha W_1(z)\right] z^{-1} + \beta W_1(z)$$

$$Y(z) = \left\{\left[z^{-1} - \alpha\right] z^{-1} + \beta\right\} W_1(z)$$

$$Y(z) = KX(z)\frac{z^{-2} - \alpha z^{-1} + \beta}{1 - \alpha z^{-1} + \beta z^{-2}}$$

$$\boxed{\frac{Y(z)}{X(z)} = \frac{K\left[\beta - \alpha z^{-1} + z^{-2}\right]}{1 - \alpha z^{-1} + \beta z^{-2}}}$$

(b)

$$H(z) = K\frac{\beta - \alpha z^{-1} + z^{-2}}{1 - \alpha z^{-1} + \beta z^{-2}}$$

$$H(\theta)|_{\theta=0} = K\frac{\beta - \alpha + 1}{1 - \alpha + \beta} = 1 \qquad \text{(given)}$$

$$\boxed{K = \frac{1 - \alpha + \beta}{\beta - \alpha + 1} = 1}$$

(c)

$$H(\theta) = K\frac{\beta - \alpha e^{-j\theta} + e^{-j2\theta}}{1 - \alpha e^{-j\theta} + \beta e^{-j2\theta}}, \qquad H(\theta)|_{\theta=\pi} = 1 \quad \text{(given)}$$

$$1 = K\frac{\beta - \alpha e^{-j\pi} + e^{-j2\pi}}{1 - \alpha e^{-j\pi} + \beta e^{-j2\pi}} = K\frac{\beta - \alpha + 1}{1 - \alpha + \beta}$$

$$\therefore \quad \boxed{K = 1}$$

(d)

$$H(0) = 1$$

$$H(\pi) = 1$$

$H(z)$ is an all-pass function with a constant magnitude at all values of θ.

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20. (a) Calculate the magnitude and phase responses of the following FIR filter:

$$y[n] = \frac{1}{4}[x[n] - 2x[n-1] + x[n-2]]$$

(b) Sketch the magnitude and phase responses.

(c) Show that the filter has a constant group delay and determine its value.

(a)

$$H(z) = \frac{1}{4}\left(1 - 2z^{-1} + z^{-2}\right)$$

$$H(\theta) = \frac{1}{4}\left(1 - 2e^{-j\theta} + e^{-j2\theta}\right)$$

$$= \frac{1}{4}e^{-j\theta}\left(e^{j\theta} - 2 + e^{-j\theta}\right)$$

$$= \frac{1}{4}e^{-j\theta}(2\cos\theta - 2)$$

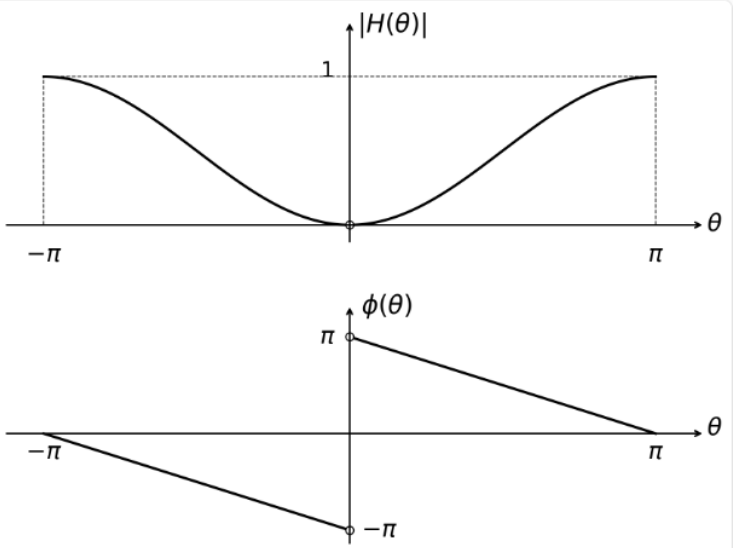
$$= e^{-j\theta}\underbrace{\frac{1}{2}(\cos\theta - 1)}_{B(\theta)}$$

$$|B(\theta)| = \left|\frac{1}{2}(\cos\theta - 1)\right|$$

$B(\theta)$ is always negative.

$$\phi(\theta) = -\theta \pm \pi, \qquad B(\theta) \le 0$$

(b)



(c) Group delay

$$\tau_g = -\frac{d\phi(\theta)}{d\theta}$$

$$= -\frac{d}{d\theta}(-\theta \pm \pi)$$

$$= 1 \qquad \text{Constant.}$$